

The GameStop Squeeze: When Momentum Models Broke

During January 2021, short-momentum strategies failed because the most crowded trade in the market became the most dangerous one.

In January 2021 a dying mall retailer became, briefly, the most traded stock on Earth. GameStop rose ~2,700% in three weeks, a hedge fund with a decade of strong returns needed a bailout, and every systematic strategy holding “short the weak stocks” learned that its risk model had a blind spot. This case study reconstructs the squeeze from market data and examines why quantitative models — trained on decades in which shorting losers worked — broke.

PROJECT CARD

CATEGORY

Algorithmic Trading

LIBRARIES

Pandas · Matplotlib · yfinance

ASSETS

GME + XRT + ^VIX

TIMEFRAME

Oct 2020 – Mar 2021 (event window)

USE

Understanding crowding and short-squeeze risk before your systematic strategy meets one

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

plt.rcParams["figure.dpi"] = 110
```

1. The setting

By late 2020, GameStop (GME) was a consensus short: a brick-and-mortar game retailer in a download world. Reported **short interest exceeded 100% of the float** — more shares sold short than were available to trade — via rehypothecation. For a momentum or quality model, GME

scored terribly on every factor; being short was the “safe,” crowded, model-approved position.

The vulnerability hiding in that consensus: a short position has **unlimited loss** and, when everyone must exit at once, exiting *is* buying. The crowd was the risk.

2. The event

- **Jan 11–13:** GME adds board members from Chewy’s founder Ryan Cohen; the stock jumps ~60%. Retail buying accelerates, coordinated openly on r/wallstreetbets.
- **Jan 19–22:** Short-seller reports trigger not selling but *more* buying. Heavy call-option volume forces market makers to hedge by buying stock.
- **Jan 25–27:** The vertical phase — GME goes from ~\$65 to ~\$347. Melvin Capital takes a \$2.75B injection. Short interest starts collapsing as funds capitulate.
- **Jan 28:** Peak intraday ~\$483. Several brokers **restrict buying** (position-close-only), citing clearinghouse margin. The squeeze breaks.
- **Feb–Mar:** Collapse to ~\$40, then a second, smaller squeeze in late February.

3. The mechanism: two squeezes feeding each other

Short squeeze: rising price → shorts face margin calls → they buy to cover → price rises further. The exit *is* fuel.

Gamma squeeze: retail buys short-dated calls → dealers who sold the calls are short gamma and must buy stock as it rises to stay hedged → price rises → deltas rise →

dealers buy more. A mechanical accelerant layered on the behavioral one.

Neither loop cares about fundamentals. Once ignited, price becomes a function of *positioning*, not value — precisely the variable most quant models didn’t include.

4. The evidence

4.1 The data: GME, its ETF host, and the market’s fear gauge

```
TICKERS = ["GME", "XRT", "^VIX"]
START, END = "2020-10-01", "2021-03-31"

def load_prices(tickers, start, end):
    """Adjusted-close prices via yfinance."""
    import yfinance as yf
    df = yf.download(tickers, start=start, end=end, auto_adjust=True, progress=False)["Close"]
    return df[tickers].dropna()

px = load_prices(TICKERS, START, END)
rets = px.pct_change().dropna()
print(f"{len(px)} trading days, {px.index[0].date()} → {px.index[-1].date()}")
print(f"\nGME: start {px['GME'].iloc[0]:.2f}, peak {px['GME'].max():.2f} "
      f"({px['GME'].idxmax().date()}), end {px['GME'].iloc[-1]:.2f}")
```

4.2 The squeeze in one chart

Log scale — the only way to see both the base and the spike. Annotations mark the phase transitions.

```

fig, ax = plt.subplots(figsize=(11, 5.5))
ax.plot(px.index, px["GME"], color="crimson", lw=1.5)
ax.set_yscale("log")
events = {
    "2021-01-13": "Cohen board news",
    "2021-01-22": "gamma squeeze ignites",
    "2021-01-28": "peak / buying restricted",
    "2021-02-24": "second squeeze",
}
for d, label in events.items():
    d = pd.Timestamp(d)
    if d in px.index or True:
        ax.axvline(d, color="gray", ls=":", lw=1)
        ax.annotate(label, (d, px["GME"].max()*0.7), rotation=90,
                    fontsize=8, ha="right", va="top")
ax.set_title("GME adjusted close (log scale)")
ax.set_ylabel("Price ($, log)")
plt.tight_layout(); plt.show()

jan = rets.loc["2021-01", "GME"]
print(f"January 2021: {(1+jan).prod()-1:+.0%} in one month")
print(f"Biggest single days: {jan.nlargest(3).apply('{:+.0%}'.format).to_dict()}")
print(f"Worst single day: {jan.min():+.0%}")

```

4.3 Collateral damage: the ETF that couldn't help itself

GME sat inside XRT (SPDR Retail ETF). As GME went vertical, its weight in the “diversified” ETF exploded — briefly approaching ~20% — dragging a passive vehicle into the squeeze.

```

fig, ax1 = plt.subplots(figsize=(11, 4.5))
ax1.plot(px.index, px["XRT"] / px["XRT"].iloc[0] * 100, color="steelblue", label="XRT (indexed)")
ax1.set_ylabel("XRT (indexed to 100)", color="steelblue")
ax2 = ax1.twinx()
ax2.plot(px.index, px["^VIX"], color="gray", alpha=0.7, label="VIX")
ax2.set_ylabel("VIX", color="gray")
ax1.axvspan(pd.Timestamp("2021-01-22"), pd.Timestamp("2021-02-02"), color="crimson", alpha=0.10)
ax1.set_title("The spillover: XRT surges with its runaway holding; VIX wakes up")
plt.tight_layout(); plt.show()

print(f"XRT move, Jan 22-27: {px.loc['2021-01-27', 'XRT']/px.loc['2021-01-22', 'XRT']-1:+.1%}")
print(f"VIX, Jan 25-27: {px.loc['2021-01-25': '2021-01-27', '^VIX'].round(1).to_list()}")
    f" - a single stock moving the market's fear gauge")

```

4.4 What it did to a systematic short

Simulate the naive quant position: short GME with monthly rebalancing (the classic momentum/quality short book, isolated to one name). Sizing at just 2% of a book, the January move alone is catastrophic — and daily mark-to-market shows why margin forced covering *before* any month-end rebalance.

```

# short position P&L, daily compounding (mark-to-market of a static short entered Jan 4)
short_ret = -rets.loc["2021-01", "GME"]
pnl_path = (1 + short_ret).cumprod() - 1

fig, ax = plt.subplots(figsize=(10, 4))
ax.plot(pnl_path.index, pnl_path * 100, color="crimson")
ax.axhline(-100, color="black", ls="--", lw=1, label="-100% = position wiped out")
ax.set_title("P&L of a short position in GME entered Jan 4, 2021 (mark-to-market)")
ax.set_ylabel("Cumulative P&L (%)"); ax.legend()
plt.tight_layout(); plt.show()

```

```
print(f"Short P&L by Jan 27: {pnl_path.loc['2021-01-27']:+.0%}")
print("A 2% book position: ~-40% portfolio hit before risk systems could rebalance.")
print("This asymmetry – shorts grow as they lose – is the structural lesson.")
```

5. Post-mortem

What failed in the models

- **Factor models scored GME correctly and were destroyed anyway** — the signal wasn't wrong about value; it was blind to *positioning*. Short interest and borrow cost weren't inputs.
- **Risk models assumed exits exist** — liquidation models presume you can cover near current prices. In a squeeze, covering moves the price against you; the crowd exits through one door.
- **Normal-market position sizing** — a 100%+ short-interest name has a fat right tail *by construction*. Sizing it like any other short ignored that the loss distribution was not remotely log-normal (see our fat-tails work).

What survived

- **Strategies with short-interest / crowding filters** side-stepped the worst names.
- **Hard per-name loss limits** — the dumb, old-fashioned stop — beat sophisticated covariance-based risk for this event.

- **The squeeze faded** — by April GME was back below \$200 and momentum factors normalised; the *event* was survivable, the *sizing* often wasn't.

The lessons

1. **Crowding is a risk factor.** Short interest, days-to-cover and borrow cost belong in the model, not the footnotes.
2. **Shorts need asymmetric sizing** — the position grows as it hurts you; the equivalent long shrinks.
3. **Reflexivity is real:** when positioning drives price, historical covariances describe a market that no longer exists.
4. **Market structure matters** — gamma hedging turned option flow into a price accelerant; ignoring the options market meant missing half the mechanism.

Related content

- **Alpha Decay** (next) — crowding's slower cousin: everyone finding the same signal
- **Fat tails / GBM tutorial** — the distributional assumption squeezes violate
- **SMA Crossover Backtest** — where systematic discipline helps rather than hurts